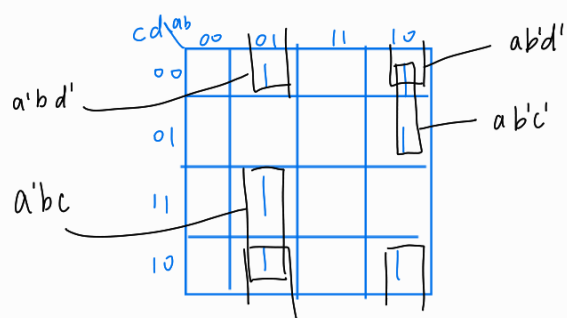
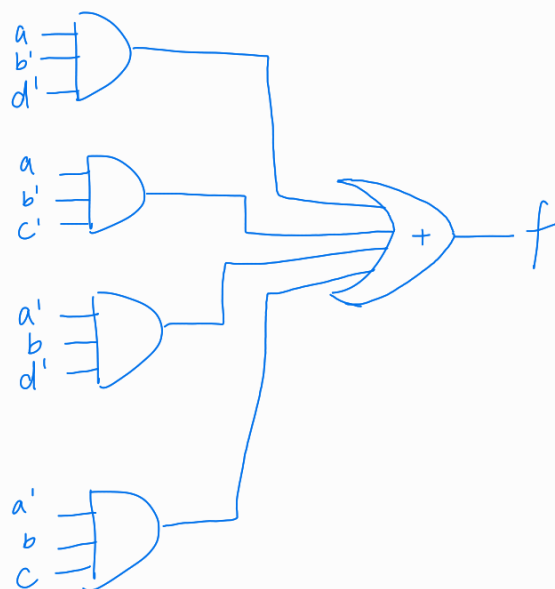




(a)

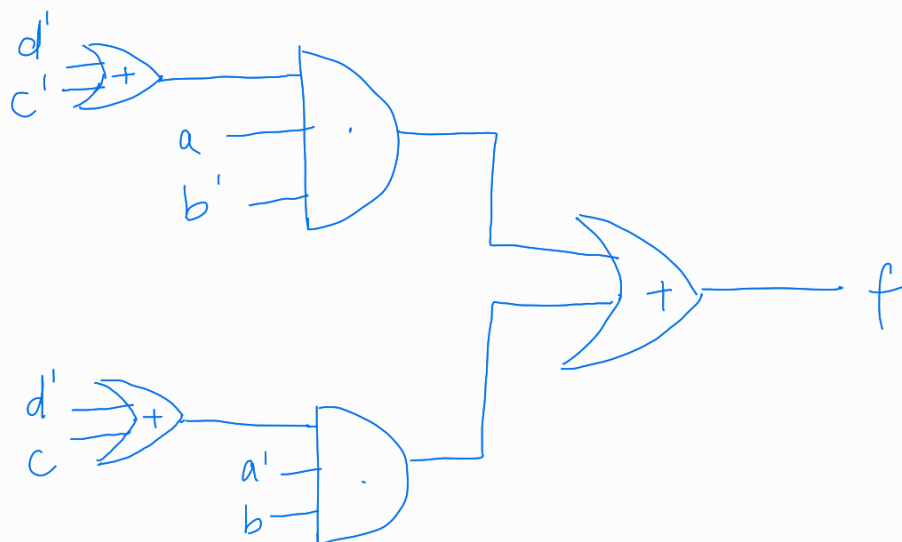
$$f = ab'd' + ab'c' + a'bd' + a'bc$$



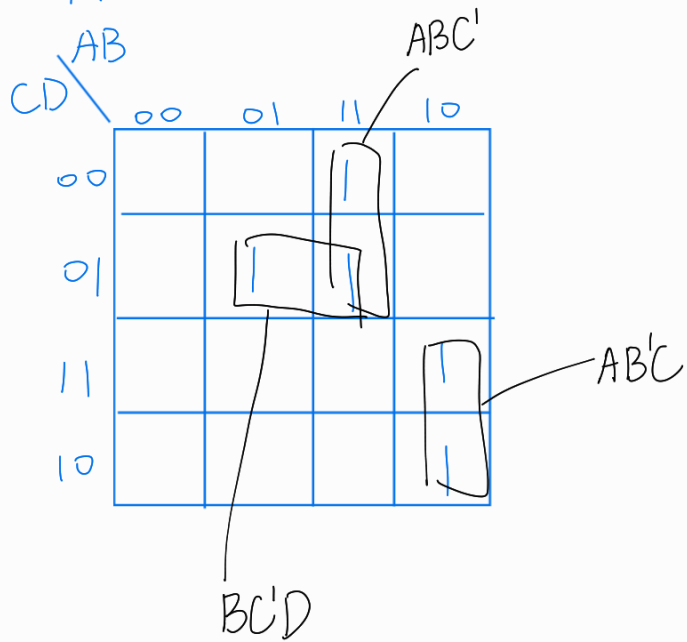
(b)

$$f = ab'd' + ab'c' + a'bd' + a'bc$$

$$= ab'(d' + c') + a'b(d' + c)$$

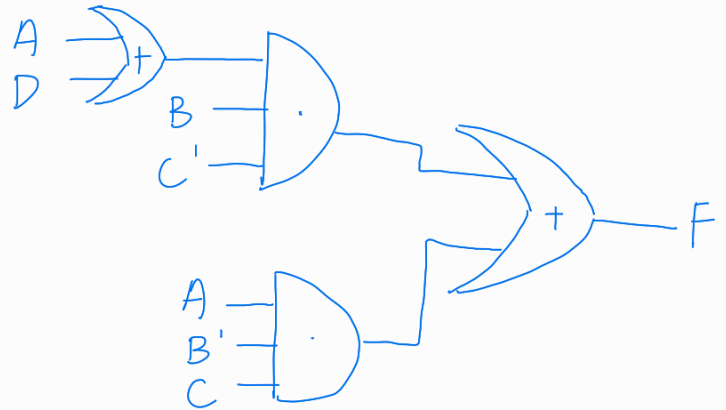


2. $F(A,B,C,D) = \sum m(5,10,11,12,13)$



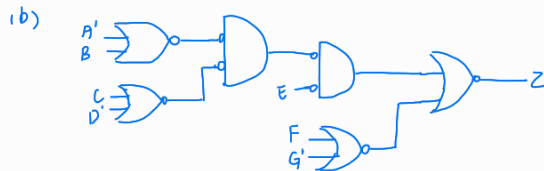
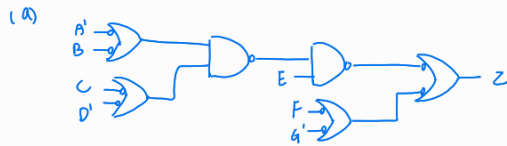
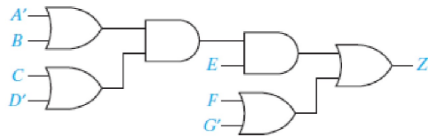
$$F = ABC' + ABC' + BC'D$$

$$= (A+D)BC' + ABC'$$



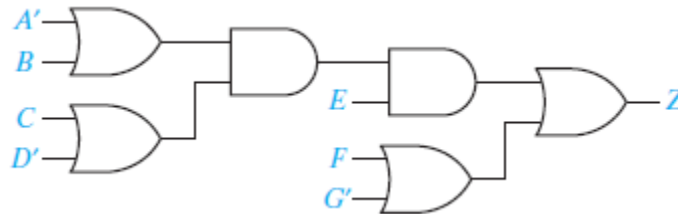
3.

3. (a) Convert the following circuit to all NAND gates, by adding bubbles and inverters where necessary.
 (b) Convert to all NOR gates (an inverter at the output is allowed).



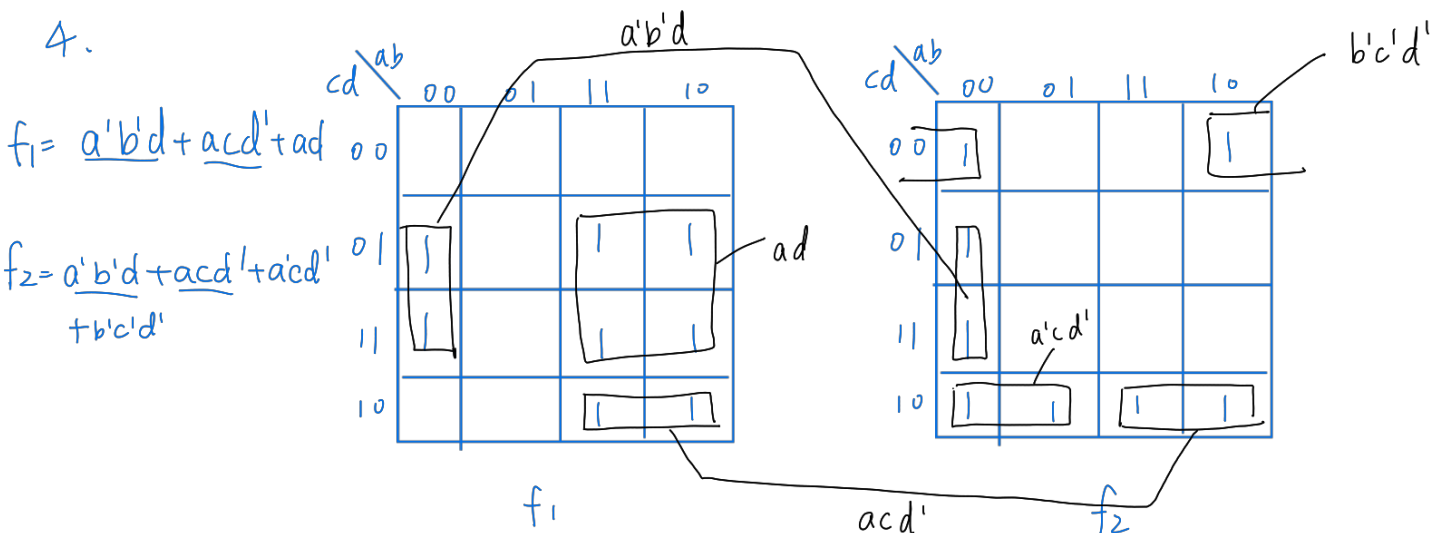
Digital Logic Design Homework 8

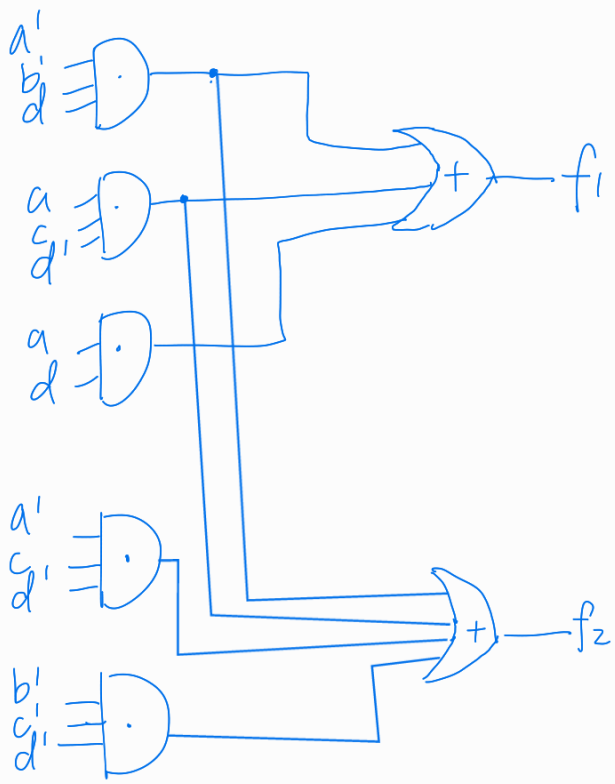
- Using AND and OR gates, find a minimum circuit to realize $f(a,b,c,d)=m_4+m_6+m_7+m_8+m_9+m_{10}$
 - using two-level logic
 - using three-level logic (12 gate inputs minimum)
- Find a minimum three-level NAND-gate circuit to realize $F(A,B,C,D)=\sum m(5,10,11,12,13)$ (four gates)
- Convert the following circuit to all NAND gates, by adding bubbles and inverters where necessary.
 - Convert to all NOR gates (an inverter at the output is allowed).



- Find a two-level, multiple-output AND-OR gate circuit to realize the following functions. Minimize the required number of gates (six gates minimum).
 $f_1 = ac + ad + b'd$ and $f_2 = a'b' + a'd' + cd'$
- Find a minimum two-level NAND-NAND circuit to realize the functions given
 $f_1 = \sum m(2,3,5,7,8,9,10,11,13,15);$
 $f_2 = \sum m(2,3,5,6,7,10,11,14,15);$
 $f_3 = \sum m(6,7,8,9,13,14,15)$
 - Find a minimum two-level NOR-NOR circuit to realize the functions given.

4.





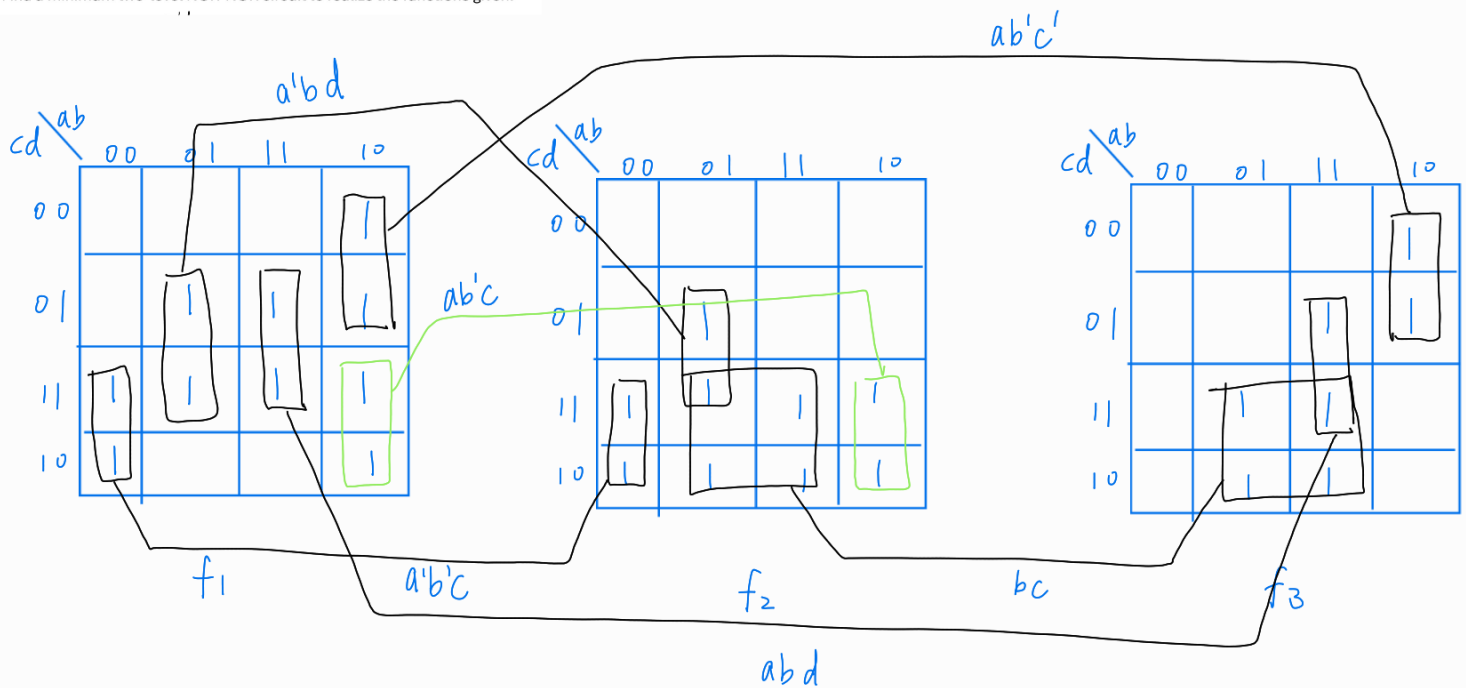
5. (a) Find a minimum two-level NAND-NAND circuit to realize the functions given

$$f_1 = \sum m(2, 3, 5, 7, 8, 9, 10, 11, 13, 15);$$

$$f_2 = \sum m(2, 3, 5, 6, 7, 10, 11, 14, 15);$$

$$f_3 = \sum m(6, 7, 8, 9, 13, 14, 15)$$

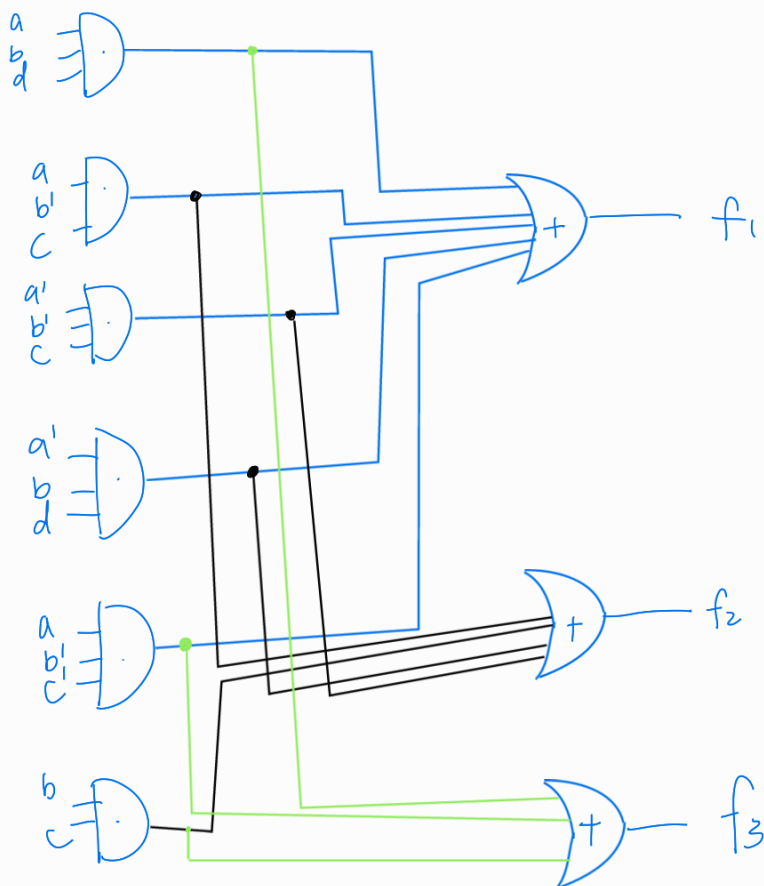
(b) Find a minimum two-level NOR-NOR circuit to realize the functions given.



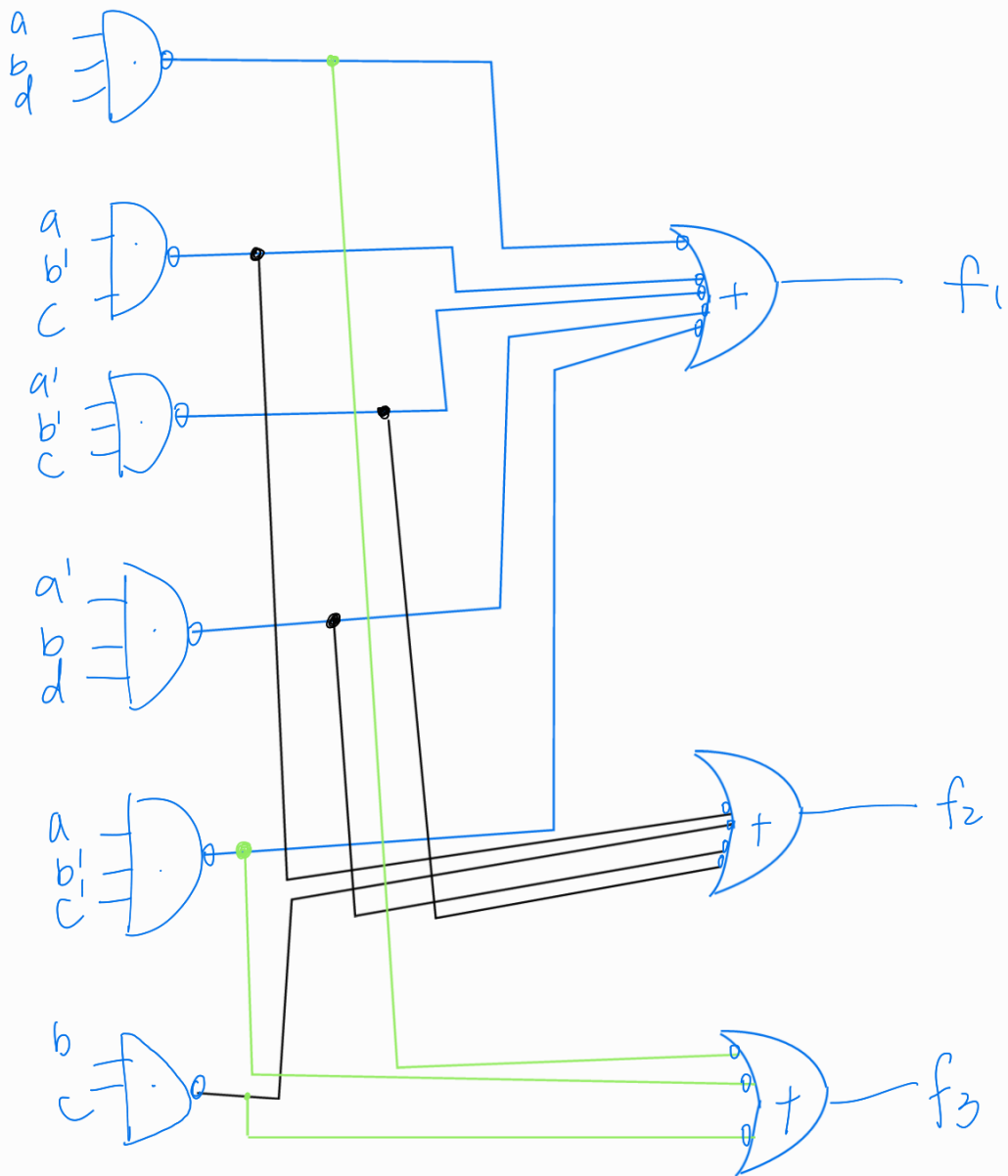
$$f_1 = \underline{a'b'c} + \underline{a'b'd} + \underline{abd} + \underline{ab'c'} + \underline{ab'c}$$

$$f_2 = \underline{a'b'c} + \underline{bc} + \underline{a'b'd} + \underline{ab'c}$$

$$f_3 = \underline{bc} + \underline{abd} + \underline{ab'c'}$$



NAND-NAND



NOR-NOR

